

Silicon lifecycle management (SLM) with in-chip monitoring

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Abstract—Increasing chip and system complexity, coupled with growing performance and reliability requirements, are driving the need for ongoing post-silicon analysis, maintenance and optimization. In-chip monitor and sensor data provide visibility into critical performance, reliability and security issues for the entirety of a chip's lifespan. Silicon lifecycle management closes the silicon loop through the analysis of the data from monitors and sensors and enables new levels of insights for both SoC teams and their customers and provide the ability to optimize operational activities at each stage of the device and system lifecycles.

I. BACKGROUND

The growing complexity of today's electronics systems is making quality and reliability increasingly difficult to achieve. This, coupled with little tolerance for performance degradation of any kind and the need to meet functional safety and security requirements, means that a new approach is needed to address how silicon-based systems are developed, operated and maintained. The potential gains and cost savings could be measured in the billions of dollars. These savings can be achieved by properly managing each phase of a chip's and system's lifecycle from development through deployment to ensure optimized results throughout. Providing access to device data throughout the entire chip life span and enabling on-going 'in life' feedback and optimization through specialized analytics will allow a more efficient and effective way to address the semiconductor-related quality and security challenges system companies face in all industries. Silicon Lifecycle Management (SLM) is a relatively new process associated with the monitoring, analysis and optimization of semiconductor devices as they are designed, manufactured, tested and deployed in end user systems. SLM is based on two underlying principles: gather as much useful data about each chip as possible and analyze that data throughout the chip's entire lifecycle to gain actionable insights to improve chip and system related activities. The Synopsys SLM Platform provides insight into critical performance, functionality, reliability, safety and security issues for the entirety of a chip's lifespan. This enables the optimization of operational activities for all participants throughout the life of the SoC.

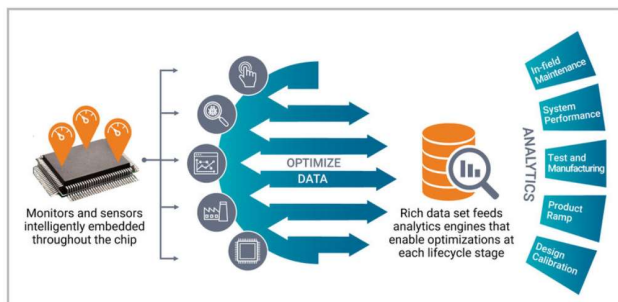


Figure 1. The SLM product lifecycle continuum from design through manufacturing to in-field

II. INTEGRATED SLM PLATFORM

The SLM Platform is tightly coupled to design implementation, signoff and test solutions, including:

- Links to design robustness analysis solution for guidance on optimal placement of the monitors and sensors.
- Automated and intelligent integration of the monitors and sensors into the RTL or gate-level design provided through the test integration solution.
- Direct links with RTL-to-GDSII solution to ensure the monitors are integrated while maintaining optimal PPA design metrics.

III. ANALYTICS

The SLM Platform includes several targeted analytics engines that operate on chip data to enable optimizations at each stage of the semiconductor lifecycle, starting with design and progressing through manufacturing, production test, bring-up and culminating with in-field operation:

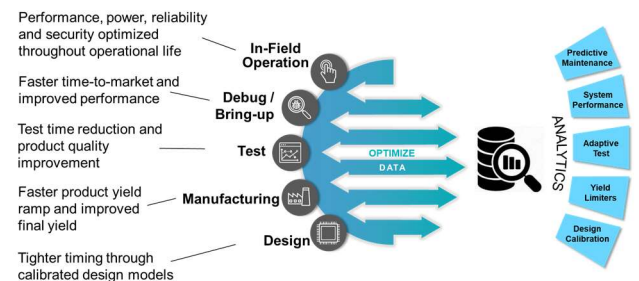


Figure 2. The SLM product lifecycle – Benefits to each SLM stage

Closing the silicon-to-design loop, minimizing required margins and enabling advanced analytics to further optimize design PPA, reliability and silicon predictability.

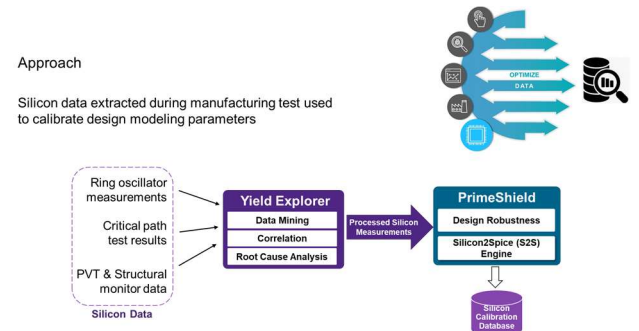


Figure 3. The SLM product lifecycle – Design Optimization loop, feedback from Silicon data back to design optimization

Semiconductor manufacturing analytics engine and the design sensitivity analysis engine use test data enhanced with monitor and sensor data to optimize manufacturing and test operational efficiencies as well as reduce design-related yield limiters.

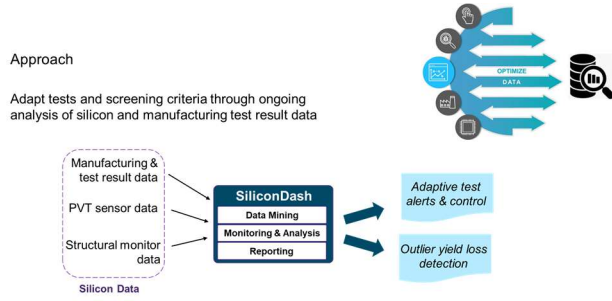


Figure 4. The SLM product lifecycle – Yield Ramp and High Volume Production Optimization

Adaptive Learning Engine and Embedded Learning Engine enable optimized bring-up and introduce self-analysis as well as safety, security and predictive maintenance capabilities.

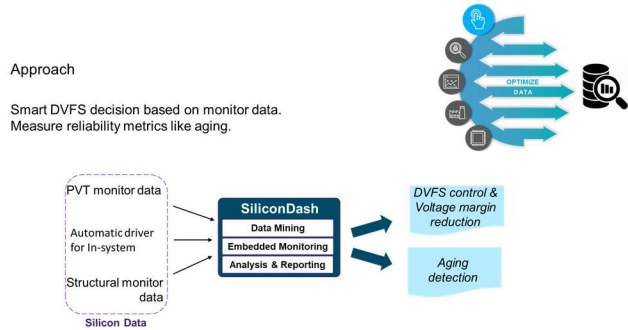


Figure 5. The SLM product lifecycle – Example of In-Field Optimization

The SLM Platform exclusively combines extensive silicon device data and targeted analytics to greatly improve how chips and systems are designed, manufactured, deployed and maintained in the field.

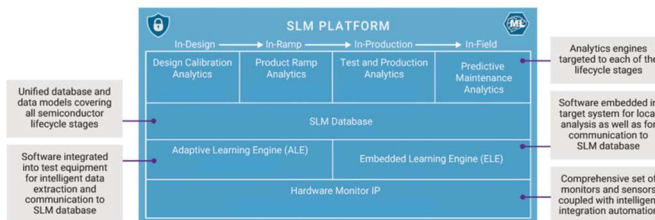


Figure 6. The Synopsys SLM platform

IV. MONITOR CLASSIFICATIONS

IC developers must face the challenges associated with the increased gate density of advanced CMOS technologies. Challenges include overheating, greater in-chip process variability and unreliable voltage supplies. Embedded in-chip monitoring, previously a nice to have item, is not an essential requirement of any advanced node chip design. They support the semiconductor design community's demands for increase device reliability and enhanced performance optimization,

helping to bring success by differentiating the customers technology. Monitor IPs falls under 7 categories Test, Structural, Process, Voltage, Temperature, Debug and Functional monitors. The table below shows the different classifications and an example under each of them and the value proposition of each. Data from multiple monitors could feed into addressing a particular SLM use case.

Type	Example	Impact
Test	Scan & BIST	Silicon bring-up, manufacturing and in-field repair
Structural	Gate delay & Path margin	Feedback to design to improve PPA, product bring-up & in-field visibility
Process	Ring oscillator	Reduce model to hardware correlation, silicon bring-up, manufacturing & in-field visibility
Voltage	DC voltage drop detector	Used in product bring-up, DVFS, detect security attacks
Temperature	Distributed thermal sensor	Used in product bring-up, DVFS, detect security attacks
Debug	Trace and status monitor	Monitors that help with product bring-up
Functional	Bus, security, safety monitors	Both for product bring-up and in-field visibility and optimization

Figure 7. SLM HW monitor types

V. MONITOR EXAMPLE

Test monitors like Scan, LBIST and MBIST are used for silicon bring-up and as screening tests for high volume manufacturing. In some vertical segments like servers and automotive, MBIST is used for in-field repair.

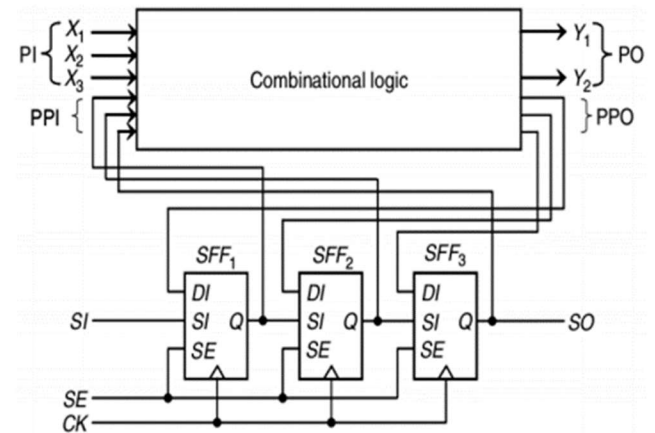


Figure 8. SLM HW monitor – Scan instrument [1]

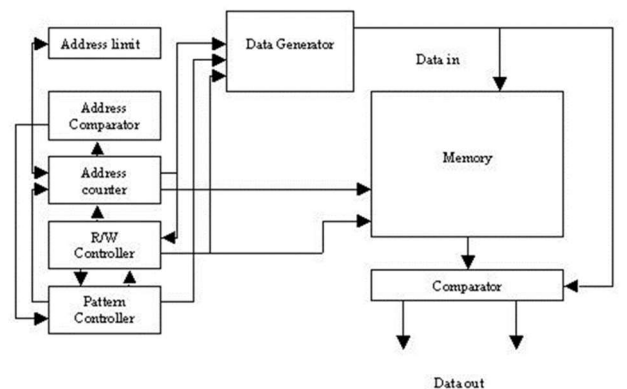


Figure 9. SLM HW monitor – Memory BIST [2]

Structural monitors like gate delay, critical path and flip-flop margin monitors are used to give feedback on design margins and to improve pre-silicon to post-silicon correlation. They can also be used to measure in-field silicon behavior over time and get feedback on attributes like aging and other reliability metrics.

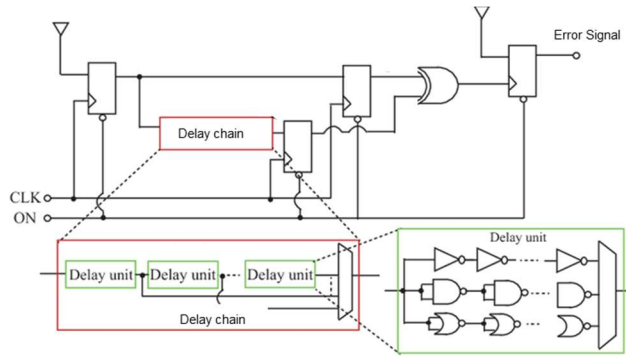


Figure 10. SLM HW monitor – Gate delay monitor [3]



Figure 11. SLM HW monitor – Path margin monitor [4]

Process monitors help with yield improvement and can act as screening for high volume manufacturing. Voltage and Temperature sensors are used to manage thermals on the SoC and are a key part of the DVFS optimization routine. They can also be used to detect Cold Boot security attacks.

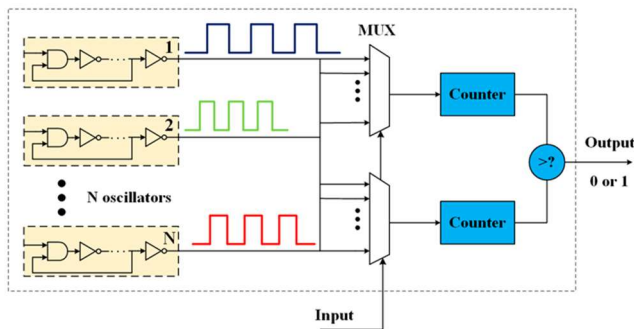


Figure 12. SLM HW monitor – Process Monitor [5]

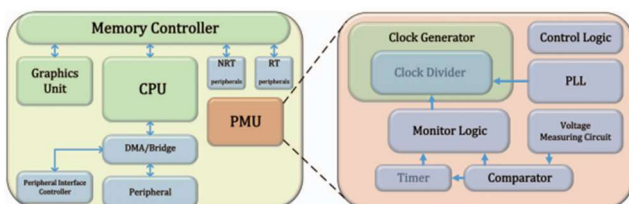


Figure 13. SLM HW monitor – Voltage and Temperature monitors [6]

Debug monitors like event detector and trace buffer are key for product debug/bring-up. Helps with debug of both HW and SW issues. In-System instruments are used for monitoring and optimization during functional operation. Some of the key use cases are for safety, security and performance optimization.

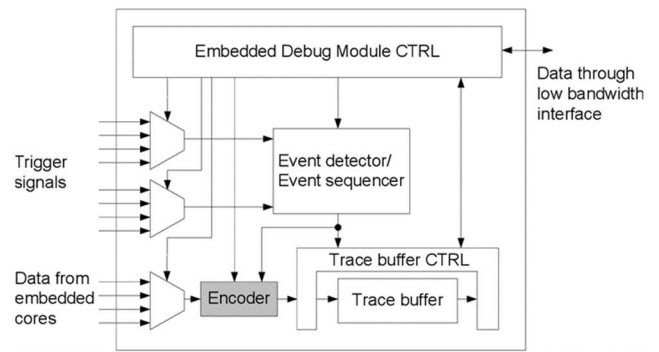


Figure 14. SLM HW monitor – Debug instruments [7]

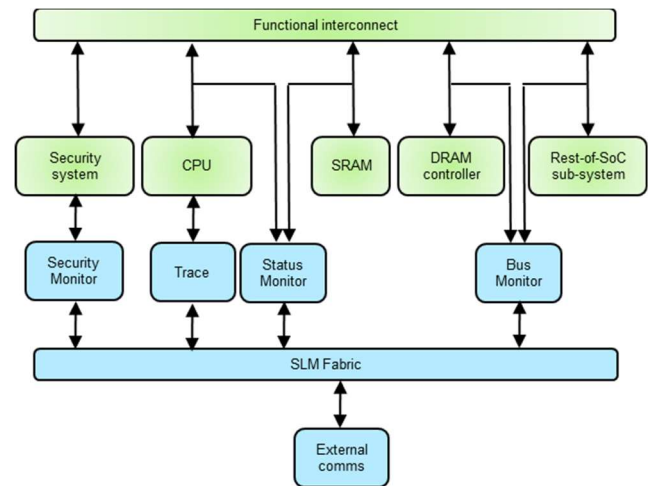


Figure 15. SLM HW monitor – Functional monitors

VI. EXAMPLE ANALYTICS

An integrated design and analytics platform would enable automated production analytics as well as detailed and comprehensive characterization of the planned on-chip sensors with minimal setup effort. The solution will be used for technology characterization/design optimization, excursion root cause identification, and for high volume production.

Analytics will be performed on wafer test data. The analytical platform will understand the intent of the sensor design including meta data that describes the sensor. During characterization and yield ramp, the analysis of PVT sensors includes general parametric analysis, identifying variability, correlating with other parametric measurements, and correlating with pre-silicon simulation data. Additionally, the path margin sensors will be used to correlate the expected margins with the silicon results across 1000s of paths.

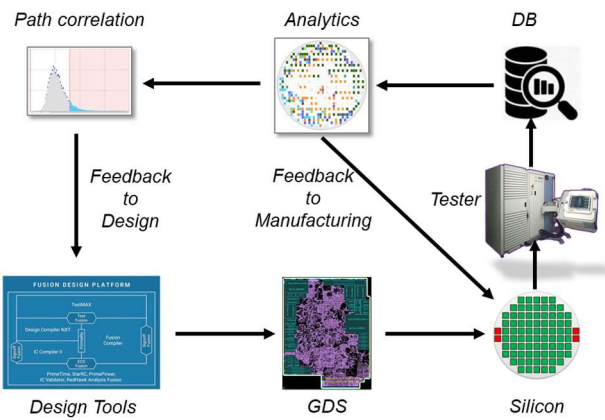


Figure 16. SLM Analytics Example – Design Feedback & High volume manufacturing

High volume manufacturing typically encompasses multiple types of data, including Wafer Sort, Final Test (FT), and System Level Test (SLT). In high volume manufacturing, the analytics will monitor for any excursion events. It will enable a new level of fine binning by grouping the devices into families of similar performance based on frequency, power, leakage current, and path margin. This helps identify any device outliers and reduces overall defective parts per million (DPPM). During final test the silicon results will again be correlated with pre-silicon simulations (HSPICE, IR drop, margin maps). Devices will be grouped into families by parametric performance and path margin to help highlight any yield fluctuations.

System level test will monitor the device performance under many different workloads after integration with prototype or system boards. Path margin and parametric analysis will be performed, allowing feedback to final test to help ensure only the most appropriate chips for the specific system and application are used. In addition, the sensors will enable the device to act as a monitor to identify and pinpoint system faults such as a failing power supply or a clock deficiency.

In the SLM platform, all the various sensor test data will be consolidated in an easy to read dashboard with ability to drilldown into any of the highlighted areas. The user can view highlighted performance characteristics of one or more products with easy notifications to identify which areas need attention. The user can drilldown from the report into many standardized analysis flows across many metrics and aggregations. A configurable analysis platform will enable new analysis methods and algorithms to be inserted into the production flow as the best practices or data needs evolve.

VII. CONCLUSIONS

In-chip monitoring is now a necessity at advanced process nodes as it enables mission critical management of increasingly variable physical and functional conditions in real-time, thereby increasing performance and reliability. Combining in-chip monitors into the Synopsys SLM platform adds a new dimension of value. The rich data provided by these monitors is an essential part of fully understanding complex activities

within the chip and to derive the greatest optimizations throughout the lifecycle of the product thus providing gains and cost savings that could be measured in the billions of dollars.

VIII. ACKNOWLEDGEMENTS

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